



CAIRO UNIVERSITY RACING TEAM

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1 - Discriminative: learns the patterns in the data that help it differentiate between classes and predict the correct class like model can predict which photo is of a cat or dog,

Generative: learns how the data is generated, then can create new content similar to the training data. The model learns the differences between cats and dogs then generate a photo of a new one.

2 – Transformers are used in modern AI to understand relationships between different parts, they work mainly using self-attention, which allows each token to look at other tokens and decide which ones are the most important.

3 – supervised learning : the model is training on labeled data so it learns the patterns on the data that differentiate between the classes like classification – regression ,

Unsupervised : the model is training on unlabeled data so it learns the patterns and relationships on the data then splits the data into groups like clustering ,

Reinforcement : the model takes an action and learns by rewards(if the decision is correct) or penalties (if the decision is bad)

4 – Tokens are small units of text that LLMs process and generate , they are important because the model doesn't read and understand the text directly like us , it converts text into tokens then predicts the next token

5 – attention is a mechanism that allows the neural networks to focus more on the most important part of the inputs .

6- Prompt engineering is the process of designing clear instructions for Generative AI model to get the best result , the types include : one-shot , zero-shot , few-shot , instruction-based , role-based

7 – RAG works by retrieving relevant information from external documents or database, then gives this information to the LLM with the user question, so the model can generate an answer based on this data instead of depending only on its training data.

8 – Embedding is a way to represent text or data as numerical vectors. It is important in retrieval because similar meanings have similar vectors, so the system can find the most relevant documents to the user question.

9 – Temperature controls how random the model output is. Low temperature makes the answer more focused and consistent, while high temperature makes the answer more creative and different.

10 – Chain-of-thought prompting means asking the model to solve a problem step by step instead of giving the final answer directly. It can improve reasoning because the model breaks the problem into smaller steps.

11 – RAG uses external data without changing the model itself, like a chatbot that answers questions from company documents.

Fine-tuning means training the model more on specific data to change its behavior or make it better in a specific task, like training a model on customer service conversations.

12 – Naive RAG can be improved before retrieval and after retrieval.

Pre-retrieval improvements include better chunking, query rewriting, better embeddings and hybrid search.

Post-retrieval improvements include re-ranking the retrieved documents, removing irrelevant chunks and selecting the most useful information before sending it to the LLM.

13 – When choosing an LLM, I should consider accuracy, response time or latency, cost, context window, privacy and reasoning ability.

The best model is not always the biggest model, it depends on the requirements of the application.

14 – To check if the LLM answer is grounded, we can compare the generated answer with the provided documents and make sure that the main information in the answer exists in these documents.

We can also check answer relevance, faithfulness and citation accuracy.

15 – Computer vision can be integrated with Generative AI by allowing the model to understand images and then generate text or other content based on them.

For example, the system can take an image, understand what is inside it, then generate a description or answer questions about the image. It can be used in image captioning, OCR, visual question answering and image generation.

16 – AI Assistant for CURT

The assistant should help members from Mechanical, Electrical, Software, and Business teams ask questions about their own documents and data.

a. Architectur

User -> Logic + Access Control -> User Question -> Retrieve Documents / Data -> RAG + LLM -> Answer + Sources

Login and access control: make sure each user can only access allowed team data.

Document processing: handle files like PDF, Word, Excel, and text.

Retriever: searches for the most relevant information.

Structured data access: used when the answer needs data from databases or tables.

LLM: uses the retrieved information to generate the answer.

Citations: show where the answer came from.

b. Main Challenges

- Different file types may need different preprocessing.
- Some data may be missing, duplicated, or outdated.
- Large documents need to be split into chunks because of context limits.
- Retrieval may return irrelevant information.
- The LLM may generate answers not fully supported by the documents.

c. Possible Solutions

We can improve the system by:

- Cleaning the data and removing duplicates.
- Using good chunking with overlap.
- Using embeddings with hybrid search.
- Re-ranking retrieved results.
- Adding metadata like team name, project, and date.
- Using role-based access control for security.

d. Evaluation

I would create test questions with known answers and check:

- Did the system retrieve the correct information?
- Is the final answer correct?
- Is the answer supported by the source?
- Are the citations correct?

- Does the access control prevent unauthorized users?